Syllabus - Statistical Methods - Fall 2026

Course Information

Course Title: MATH 4720/MSSC 5720: Statistical Methods
Meeting Time: TuTh 9:30am - 10:45am
Location: Cudahy Hall 118 (Microsoft Teams - if needed)
Website: http://tinyurl.com/f26-4720

Instructor Details

- **Name:** Mehdi Maadooliat, Ph.D.
- **Office:** CU 351
- **Office Hours:** Tu 1:45pm - 3:15pm & Th 10:45am - 12:15pm in CU 351 or by e-mail

Course Description

Probability, discrete and continuous distributions. Treatment of data, point and interval estimation, hypothesis testing. Large and small sample method, regression, non-parametric methods. An introductory applications-oriented course recommended for students who wish to acquire a basic understanding of statistical methods. May not be taken for credit by those who have completed MATH 4710.

Course Topics

- Introduction to probability and probability distributions
- Sampling and descriptive measures
- Point and interval estimation
- Hypothesis testing
- Linear regression and analysis of variance
- Nonparametric methods

Learning Outcomes

By the end of the course, students will be able to:

- Apply fundamental concepts of probability to statistical problems.
- Identify and use appropriate discrete and continuous probability distributions.
- Summarize and interpret data using appropriate descriptive statistics and graphical methods.
- Describe sampling distributions and explain their role in statistical inference.
- Construct and interpret point estimates and confidence intervals for population parameters.
- Formulate and conduct hypothesis tests and interpret the results in context.
- Select and apply appropriate statistical methods for large- and small-sample problems.
- Fit and interpret simple linear regression models and assess relationships between quantitative variables.
- Apply analysis of variance to compare multiple population means.
- Apply appropriate nonparametric methods when standard parametric assumptions are not appropriate.
- Interpret statistical results in the context of real-world applications and communicate conclusions clearly.

Additional Learning Outcomes for MSSC 5720

In addition to the learning outcomes above, students enrolled in MSSC 5720 will be expected to:

- Demonstrate a deeper understanding of the theoretical foundations and assumptions underlying statistical methods.
- Analyze more complex statistical problems requiring additional mathematical and statistical reasoning.
- Compare alternative statistical procedures and justify the choice of an appropriate method.
- Critically evaluate statistical results, assumptions, and limitations.
- Communicate statistical methodology and conclusions at a level appropriate for graduate study.

Prerequisites

- MATH 1400, MATH 1410, or MATH 1450

Textbook

Title: An Introduction to Statistical Methods & Data Analysis (7th Edition)
Authors: R. Lyman Ott, Michael Longnecker
Year: 2015

Grading Breakdown

- **Homework**: 15%
- **Examinations**: 25% each (2 exams)
- **Final Exam**: 30%
- **Best Exam**: 5%

Students enrolled in MSSC 5720 will complete additional or more advanced problems on selected homework assignments and examinations. These problems will require greater theoretical depth, mathematical reasoning, and interpretation than those assigned to MATH 4720 students and are designed to assess the graduate-level learning outcomes described above.

You will **NOT** be allowed any extra credit projects to compensate for a poor average. The final grade is based on the following scale:

Grading Scale

Grade	Range
A	93.5 - 100
A-	90.0 - 93.5
B+	86.5 - 90.0
B	83.5 - 86.5
B-	80.0 - 83.5
C+	76.5 - 80.0
C	73.5 - 76.5
C-	70.0 - 73.5
D+	66.5 - 70.0
D	63.5 - 66.5
D-	60.0 - 63.5
F	below 60.0

Note: No late homework will be accepted, and no make-up work will be allowed unless in the case of an emergency. Submit incomplete work if necessary, but ensure scanned PDFs are legible before submission.

Exam Dates

- **Midterms**: Oct. 1st, Nov. 12th (70 minutes long)
- **Final Exam**: Dec 16th from 8:00 am to 10:00 am

Homework Policy

Homework is required for a better understanding of the material and must be submitted to D2L. No late homework will be accepted, and two homework assignments will be dropped. It is recommended to submit even incomplete work. You can scan your homework and submit it as a PDF, but ensure it is readable.

Generative AI

Generative AI tools such as ChatGPT may be used for clarification, coding assistance, and take-home work unless otherwise stated. Students remain fully responsible for the correctness and interpretation of all submitted work and must disclose substantive use of generative AI. Generative AI is not permitted during the in-class exam.

Attendance Policy

Attendance is required, and the College of Arts and Sciences attendance policy will be observed.

Make-up Policy

There will be no make-up exams or homework unless there is an emergency.

Honor Code

Marquette University Honor Code:
"I recognize the importance of personal integrity in all aspects of life and work. I commit myself to truthfulness, honor, and responsibility, by which I earn the respect of others. I support the development of good character and commit myself to uphold the highest standards of academic integrity."

Important dates

- **Monday, August 31:** Classes begin
- **Tuesday, September 8:** Drop/add ends
- **Thursday - Friday, October 22 - 23**: Fall Break
- **Friday, November 20:** Last day to withdraw with W
- **Wednesday - Sunday, November 25 - 29**: Thanksgiving Break
- **Saturday, December 12**: Classes end
- **Wednesday, December 16th, 8:00 am - 10:00 am**: Final Exam

For more important dates, see the full MU Academic Calendar.

Important Note

The syllabus may be modified throughout the course. Any substantial modifications will result in a reissued syllabus.